Development of a Web-Based Attendance Monitoring System for ICT Students at the Philippine Women's University

A Research Paper Presented to the Faculty of the

Department of Senior High School

Philippine Women's University Career Development and

Continuing Education Center Sta. Cruz, Laguna Branch

Partial Fulfillment of the Requirements for the Subject Practical Research II

Camero, Jacob S.

Cabriole, Nico Paulo

Salaria, Khaiyen Alexander R.

(2025)

Chapter I

THE PROBLEM AND ITS BACKGROUND

Introduction

In recent years, students prefer the ICT track because of its wide-ranging opportunities and increasing industry demand. Despite the growth of ICT tracks, staff are still able to manage and monitor student attendance effectively. A number of schools today still rely on manual or semi-manual methods like paper sign-ins or spreadsheets to track student attendance. Moreover, these techniques take an excessive amount of effort, have mistakes, and do not design well for mass students.

When some absent students are marked present, the attendance becomes incorrect, and you may have an unfair issue. In the end, this negatively impacts class engagement and may render an institutional record invalid. The limitations of the manual methods are more evident in a larger class size situation. "A necessary need to get faster and reliable tracking".

To help with these issues, the researchers have proposed a web platform for attendance monitoring. The project is an automatic attendance management system which helps in saving time and increasing efficiency of the management system. Additionally, it intends to lessen the administrative workload for teachers. Using the database all the attendance taking and monitoring of attendance will be automated.Thus, the attendance systems will become more accurate, efficient and transparent to users.

Furthermore, the System is designed to maintain digital records, which may be used for safety and administrative monitoring from time to time. An automatic notification may alert a parent when a student’s attendance is recorded at the school.

This study aims to help the attendance management of the ICT students of the Philippine Women's University in this day and age.

Background of the Study

Thirty-five schools and districts have successfully implemented BeaconUp's technology to combat low employee turnover, low happiness, low engagement to name a few, in recent years. These establishments vary from New England's small schools to California's biggest schools. The software developed by BeaconUp is now live and automated and used by the staff to measure employee happiness and engagement and improve upon it.

Conventional attendance tracking methods like a sign-in sheets on paper and spreadsheet entry put the institution at certain risks. Potential risks comprise record input issues, delayed reporting, staff workload, limited insight into student attendance patterns on the ground, etc. As class strength increases, most of these limitations get aggravated. The ability of institutional data to provide the reliability of successful academic intervention will be affected. Accurate and efficient attendance tracking is necessary to ensure academic integrity.

As the ICT sector is growing rapidly, higher education institutions are facing the mounting challenge of implementing modern monitoring techniques. ICT students are normally required to attend classes using the latest digital technologies, but the processes for managing student attendance in some cases have not kept pace with technological advances. The daily administrative university monitoring use traditional techniques. As a result, there is a gap in digital skills promoted by the curriculum.

An online attendance tracking system is the best solution to all of these problems. These systems enable recording of information instantly, storing of data